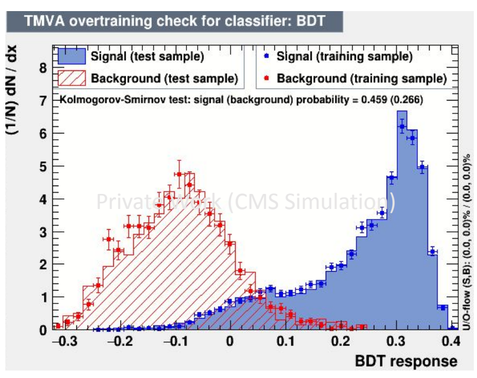
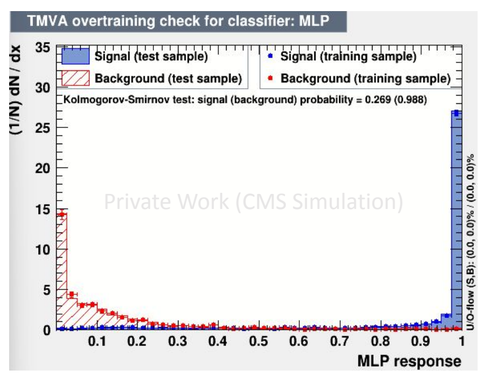
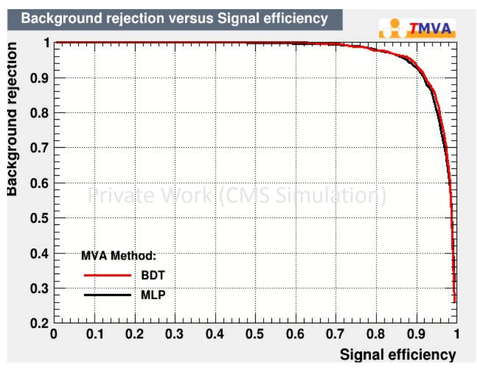
An Analysis of Multivariate Lepton Jet Recognition

Sreevardhan Atyam, Peter Dong
Illinois Mathematics and Science Academy, Fermilab







Figures 1-3: Overtraining checks for both Multi-Layer Perceptrons, and Boosted Decision Trees, BDT having generally better training; however, efficiency is superior in BDT, according to ROC curve

ABSTRACT

Dark photons are useful indicators to explain important phenomena beyond the Standard Model of particle physics, namely in the context of many experiments. Dark photons can interact with Standard Model particles through a process called kinetic mixing, which allows them to decay into Standard Model leptons, from which lepton jets are produced. However, classification of such particles becomes challenging when introducing background effects, systematic errors, and cut values that are difficult to calibrate. Therefore, we propose a methodology in which a multivariate analysis is used to produce a neural network that differentiates true signal events from fakes. Machine learning analyses can aid in fixing this issue; moreover, several neural net methods for cut efficiency and optimal cut values were employed: specifically, Boosted Decision Trees (BDT), Multi-Layer Perceptrons (MLP), and Deep Neural Networks (DNN)'s. In summary, the most successful method of separation was through the use of BDT, where we observed optimal signal acceptance and through efficiency ROC curves; we also concluded that the current efficiency for our BDT experiment was maximized at a higher degree than our MLP experiment. MLPs also had remarkably similar background rejection and signal efficiency, but were generally outperformed by BDTs.

INTRODUCTION

- To predict the best indicators that positively contribute to signal efficiency, we select *input variables* to observe signal and background classification trends
- From these trends we can recognize optimal input variables by recognizing distinct background and signal separation regions
- Input variables can be used in several ML models, more notably in BDTs and MLPs, the main foci of this analysis
- The combined output of each input variable, can show optimal cut values that can be verified and used in reference for future experiments
- The addition and deletion of certain variables *might* increase efficiency, but redundancies should be avoided

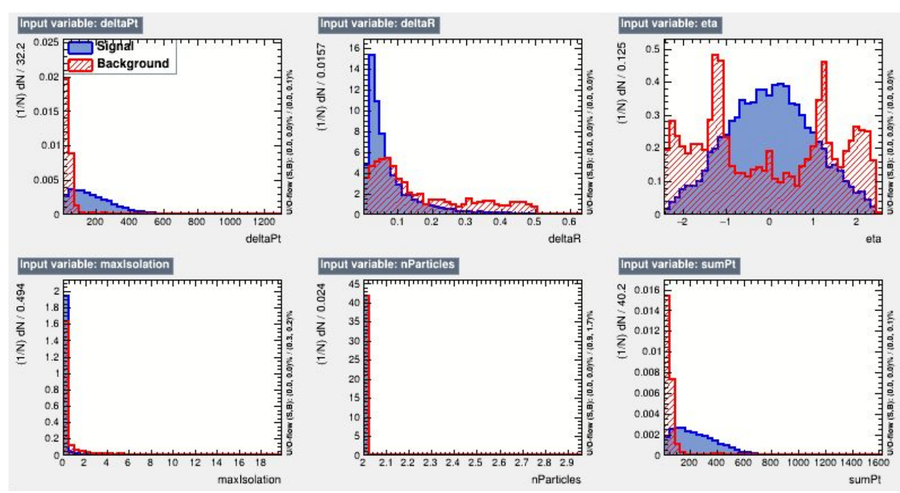


Figure 4: Shows signal and background distributions of finalized input variables. *Quality* of variable is characterized by graph peaks, and distinct signal separation

METHODOLOGY

- For each of the ML model analyses, the framework splits into five general file structures:
  - TreeMakerModule initializes and fills necessary trees for computation
  - SpecialVariableModule serves as the "appender" of variables to dataloaders, trees, factories etc.
  - MLCalculator implementing what MLStrip calculates for pre-existing sample variable values, from MC samples
  - MLTrain to train the network, using either BDTs or MLPs
  - RunAnalyzer to cross-reference network output with MC ROOT output
- Emphasizing module separation allowed for easy variable manipulation, and easier error tracing in our framework; variable and structure introduction can now be purely done in SpecialVariableModule
- We finalized on using number of particles, isolation, eta, cone size, and total/change in pT variables due to their high signal-background separation, noted by signal peaks.

CONCLUSIONS

- Through BDT efficiency ROC curves, we concluded that the sensitivity for our experiment was maximized at 28.99, with around 93% background rejection and 92% signal efficiency
- We also concluded that the current optimal efficiency for our experiment was maximized with BDT methodology. MLPs also had remarkably similar background rejection and signal efficiency, but were generally outperformed by BDTs.
- Duplicated and unexpected variable usage can cause significant computation errors, specifically because of incorrect matrix diagonalization
- Past results in our experiment reflect similar fake jet ratios for both background and signal were refuted, but due to the low overall background representation of ~20%, these conclusions support that multi-flavor lepton jets are not a good candidate for a control sample or fake estimate. Further TMVA analysis can aid in fixing this issue

REFERENCES

Cheung, C., Ruderman, J. T., Wang, L.-T., & Yavin, I. (2010). Lepton jets in (supersymmetric) electroweak processes. Journal of High Energy Physics, 2010(4). https://doi.org/10.1007/jhep04(2010)116

Sirunyan, A. M., Tumasyan, A., Adam, W., Bergauer, T., Dragicevic, M., Erö, J., Valle, A. E. D., Frühwirth, R., Jeitler, M., Krammer, N., Lechner, L., Liko, D., Madlener, T., Mikulec, I., Pitters, F. M., Rad, N., Schieck, J., Schöfbeck, R., Spanring, M., & Templ, S. (2021). Search for dark matter produced in association with a leptonically decaying Z boson in proton–proton collisions at sqrt(s)=13 TeV. The European Physical Journal C, 81(1). https://doi.org/10.1140/epjc/s10052-020-08739-5

Guglielmi, V. (2023). Machine learning approaches for parameter reweighting in MC samples of top quark production in CMS. https://arxiv.org/pdf/2312.00505.pdf

Backes, M., Butter, A., Dunford, M., & Malaescu, B. (2023). Event-by-event Comparison between Machine-Learning– and Transfer-Matrix–based Unfolding Methods. https://arxiv.org/pdf/2310.17037.pdf